Salvatore Bottiglieri

Software Engineer · Agent Infrastructure

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I build the systems that agents run in (runtimes, orchestration, tool boundaries, evaluation). Two years of shipping agents in production and five years of backend and distributed systems engineering. Three years fully remote.

EXPERIENCE

Infocube Naples, Italy

Software AI Engineer *Nov 2024 – Present*

Agents in production: Built and operate three agents that automate a security workflow end to end for a global enterprise video platform: findings triage across SAST, DAST and SCA scans; PDF reports turned into Jira tickets; tickets turned into proposed code fixes. Deployed on AWS Bedrock AgentCore and ECS.

MCP servers: Wrote and self-deployed the Jira and Confluence Model Context Protocol servers those agents run on, keeping domain logic in the server so each agent stays a thin client.

Cloud-agnostic agent library: Built the shared internal library behind the company's agents, giving vendor-agnostic orchestration and one deployment path across clouds.

Multi-agent generative UI: Built a LangGraph multi-agent system that turns natural-language constraints into custom UI components, now used for internal frontend prototyping.

Concurrency: Rewrote the core processing engine around async patterns and **halved end-to-end processing time**, with lower infrastructure cost.

AIRED Remote (3 years)

Software Engineer (Core Infrastructure) *Sept 2021 – Oct 2024*

Search: Built a hybrid search engine (sparse retrieval plus dense vectors), measurably improving relevance over the previous ranker.

CI/CD: Rebuilt the deployment lifecycle as a multi-branch pipeline, cutting time-to-production from 4 hours to **15 minutes** and enabling daily releases.

Applied ML: Built and deployed a binary classifier for medical diagnostics (AUC 0.98); peer-reviewed and published (see below).

OPEN SOURCE & SYSTEMS

Cortex — proactive agent runtime github.com/salvatorebottiglieri/botti-agent

Python, personal project *2026*

Agent runtime designed to act, not only to answer: the plan–act–verify loop can be entered by a background goal instead of a user turn; in-flight goals are persisted and resumed automatically after a restart; distributed MQTT sensor agents feed salience-filtered world events into the same loop. Also a testbed for non-standard tool-calling patterns, with a custom evaluation harness. 18 architecture decision records; CI enforces mypy and a coverage gate.

agentic-dev-kit — coding-agent guardrails github.com/salvatorebottiglieri/agentic-dev-kit

Python, open source (MIT) *2026*

Ran a pre-registered A/B eval of a coding agent on tasks drawn from production repositories (2 tasks × 5 replicas per variant). Round 1 showed that prompt-level rules changed the agent's form, not its defect rate. Replaced them with a pre-commit gate that computes the facts (skipped tests, stale deploy pins) instead of trusting the agent: defects reached the reviewer in 10/10 runs vs 2/10 for the baseline, with 0 bypasses, 0 false alarms and +14–26% cost, within a 30% budget fixed in advance. Shipped in v0.2.0.

EDUCATION

Università degli Studi di Napoli Federico II Naples, Italy

M.Sc. in Computer Science *Dec 2023*

Focus: Computational Complexity, Advanced Neural Architectures, Distributed Database Systems.

STACK

Languages: Python, Java, TypeScript, SQL, C.

Agents & LLMs: MCP, LangGraph, LangChain, tool calling, agent evaluation, RAG, PyTorch, Transformers.

Infrastructure: AWS (Bedrock AgentCore, ECS, Lambda), Azure, GCP, Docker, Kubernetes, CI/CD (Jenkins/GitHub Actions), Keycloak.

Data: Qdrant, PostgreSQL, SQLite; hybrid sparse-dense retrieval.

SELECTED PUBLICATIONS

MDPI (Journal): *Machine Learning as a Support for the Diagnosis of Type 2 Diabetes* (April 2023).

ICTS4eHealth: *Generative AI and Emotional Health: Innovations with Haystack* (June 2024).